Information Field Theory

Advanced Neuroscience

IFT-Neuroscience v1.0

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A detailed neuroscientific extension of Information Field Theory (IFT) is presented, connecting the consciousness functional Φ_{IFT} with concrete experimental protocols. Competitive experiments are designed to differentiate IFT from IIT and GNWT, based on the prediction of posterior predominance for conscious content (IFT) versus frontal ignition (GNWT) or global integration (IIT). Quantitative predictions are derived for anesthesia (differential reduction of Φ_{IFT} vs. $\tau\Xi$ depending on the anesthetic agent), sleep (REM: elevated Φ_{IFT} with reduced frontal control), and psychedelics (increase in Fisher curvature). Real-time measurement protocols for Φ_{IFT} using high-density EEG and simultaneous fMRI are proposed. IFT-Neuroscience provides a falsifiable and quantitative framework for the neuroscience of consciousness.

Contents

Chapter 1

Introduction: IFT in Neuroscience

1.1 From Theoretical Framework to Experimental Validation

The IFT-Consciousness document established the theoretical framework: consciousness as a regime of informational self-presence with the dual criterion $\Phi_{\text{IFT}} > \Phi_c \wedge \tau\Xi > K$. This document provides the detailed experimental protocols to validate or refute these predictions in the laboratory.

1.2 Objectives

1. Design experiments that differentiate IFT from IIT and GNWT.
2. Derive quantitative predictions for altered states of consciousness.
3. Propose real-time measurement protocols for Φ_{IFT}.
4. Establish operationalizable refutation criteria.

Chapter 2

Differentiable Predictions from IIT and GNWT

2.1 Three Theories, Three Predictions

Phenomenon	IIT	GNWT	IFT
Conscious content	Global integration	Frontal ignition	Posterior predominance
Unconscious activity	Low Φ	No ignition	Subthreshold Φ_{IFT}
Cerebellum	Low integration	No ignition	Low I_{causal}
REM sleep	High Φ	Partial ignition	High Φ_{IFT}, reduced $\tau\Xi$
Anesthesia	Collapsed Φ	Blocked ignition	Φ_{IFT} or $\tau\Xi$ drop

2.2 Competitive Experimental Design

2.2.1 Experiment 1: Localization of Conscious Content

- **Task**: Visual perception with report (e.g., masked stimulus).
- **Measurement**: Simultaneous EEG/fMRI during conscious vs. unconscious trials.
- **IFT Prediction**: Stimulus decodability will be higher in posterior cortex (occipital, temporal) than in prefrontal during conscious trials.
- **GNWT Prediction**: Prefrontal ignition will be necessary and sufficient for conscious report.
- **IIT Prediction**: Global integration (measured by Φ^{IIT}) will distinguish conscious trials.
- **Analysis**: Compare regional Φ_{IFT} (posterior vs. frontal) with Φ^{IIT} and GNWT ignition indices.

2.2.2 Experiment 2: Cerebellum and Consciousness

- **Task**: Motor learning (conscious vs. implicit).
- **Measurement**: High-resolution fMRI in cerebellum and cortex.
- **IFT Prediction**: $\Phi_{\mathrm{IFT}}^{\mathrm{cerebellum}} < \Phi_{\mathrm{IFT}}^{\mathrm{cortex}}$ even during complex tasks.

Chapter 3

Predictions for Altered States of Consciousness

3.1 General Anesthesia

IFT predicts that different anesthetics affect different components of the dual criterion:

Anesthetic	**Primary Effect**	**IFT Mechanism**
Propofol	Potentiates GABA-A	Reduces Φ_{IFT} (cortical integration)
Ketamine	Blocks NMDA	Disorganizes I_{causal} without collapsing Φ_{IFT}
Sevoflurane	Mixed	Reduces $\tau\Xi$ (arousal) and Φ_{IFT}
Dexmedetomidine	α_2 agonist	Reduces τ (arousal), similar to NREM sleep

Differentiable prediction: Ketamine produces dissociation (disorganized conscious experience) while propofol produces unconsciousness. IFT predicts that Φ_{IFT} can remain elevated with ketamine but collapses with propofol.

3.2 REM Sleep

IFT predicts:

- $\Phi_{\text{IFT}} \geq \Phi_c$ (rich oneiric experience, elevated posterior and limbic activity).
- Reduced $\tau\Xi$ compared to wakefulness (less temporal coherence, fragmented narrative).
- Reduced frontal control (lower prefrontal contribution to Φ_{IFT}).

3.3 Psychedelics (Psilocybin, LSD)

IFT predicts:

- $\text{Tr}(\kappa_F) \uparrow$ (increase in informational curvature, greater phenomenal richness).
- I_{causal} may become disorganized (loss of hierarchical structure).
- $\tau\Xi$ may increase or decrease depending on dose (altered coherence window).

3.4 Deep Meditation

IFT predicts:

- $\tau \uparrow$ (greater temporal stability, reduction of mental noise).
- Reorganization of ρ_{base} (change in the homeostatic reference state).
- Ξ may decrease (less reactivity to stimuli, equanimity).
- Φ_{IFT} remains elevated but with reduced content (objectless consciousness).

Chapter 4

Real-Time Measurement Protocol for Φ_{IFT}

4.1 Reconstruction of $\rho(x,t)$

Use high-density EEG (256 electrodes) combined with simultaneous fMRI:

$$\rho_i(t) = \frac{P_{\text{EEG}}(x_i,t) + \alpha \cdot \text{BOLD}(x_i,t)}{\sum_j (P_{\text{EEG}}(x_j,t) + \alpha \cdot \text{BOLD}(x_j,t))}, \tag{4.1}$$

where α is a multimodal calibration parameter.

4.2 Calculation of Ξ

$$\Xi_i(t) = \left| \sum_j A_{ij} \left[\ln \rho_j(t) - \ln \rho_i(t) \right] \right|, \tag{4.2}$$

with A_{ij} the structural connectivity matrix (DTI) or functional connectivity matrix (partial correlation).

4.3 Calculation of τ

Exponential fit of the temporal autocorrelation:

$$C_i(\Delta t) = \langle \rho_i(t)\rho_i(t+\Delta t) \rangle \sim e^{-\Delta t/\tau_i}. \tag{4.3}$$

4.4 Calculation of I_{causal}

Using transfer entropy among the 68 Desikan–Killiany regions:

$$TE(i \to j) = H(x_j^{t+1} | x_j^t) - H(x_j^{t+1} | x_j^t, x_i^t). \tag{4.4}$$

Greedy bipartition to approximate the maximum over partitions.

4.5 Assembly of $\Phi_{\text{IFT}}(t)$

$$\Phi_{\text{IFT}}(t) = \sum_i \Delta V_i \left[\alpha_I \rho_i \ln \frac{\rho_i}{\rho_i^{\text{base}}} + \alpha_F \ell_F^2 \operatorname{Tr}(\kappa_F)_i \right] + \alpha_C I_{\text{causal}}. \tag{4.5}$$

4.6 Real-Time Visualization

- **Panel 1**: Topographic map of regional Φ_{IFT} (posterior vs. frontal cortex).
- **Panel 2**: Temporal evolution of $\Phi_{\text{IFT}}(t)$ and $\tau(t)\Xi(t)$.
- **Panel 3**: Conscious state indicator: green ($\Phi_{\text{IFT}} > \Phi_c \wedge \tau\Xi > K$), yellow (threshold), red (subthreshold).

Chapter 5

Operationalized Refutation Criteria

1. **Refutation by localization**: If in Experiment 1 frontal decodability consistently exceeds posterior for conscious content.
2. **Refutation by anesthesia**: If $\Phi_{\mathrm{IFT}} > \Phi_c$ during deep propofol anesthesia in the absence of reported experience.
3. **Refutation by ketamine**: If ketamine does not produce elevation of Φ_{IFT} relative to propofol in the same subject.
4. **Refutation by REM sleep**: If Φ_{IFT} during REM sleep is indistinguishable from wakefulness.
5. **Refutation by psychedelics**: If $\mathrm{Tr}(\kappa_F)$ does not increase significantly under psychedelics.
6. **Refutation by cerebellum**: If $\Phi_{\mathrm{IFT}}^{\mathrm{cerebellum}} \approx \Phi_{\mathrm{IFT}}^{\mathrm{cortex}}$ during complex conscious tasks.

Chapter 6

Research Program

6.1 Phase 1: Calibration (Year 1)

- Determine Φ_c and K via a study with $N = 100$ participants in wakefulness, sleep, and light anesthesia.
- Validate ρ estimators against ground truth (simultaneous EEG-fMRI).
- Publish preregistered protocol.

6.2 Phase 2: Validation (Years 2–3)

- Competitive study of IFT vs. IIT vs. GNWT with $N = 256$.
- Blind analysis by an independent team.
- Success criterion: $r(\Phi_{\mathrm{IFT}}, \text{consciousness}) > 0.7$.

6.3 Phase 3: Clinical Applications (Years 3–5)

- Diagnosis of covert consciousness in non-responsive patients.
- Monitoring of anesthetic depth.
- Assessment of minimally conscious states.

Chapter 7

Master Formula of IFT-Neuroscience

$$\rho_i(t) = \frac{P_i(t) + \epsilon}{\sum_j (P_j(t) + \epsilon)} \quad \text{(Density from data)}, \tag{7.1}$$

$$\Phi_{\text{IFT}}(t) = \sum_i \Delta V_i \left[\alpha_I \rho_i \ln \frac{\rho_i}{\rho_i^{\text{base}}} + \alpha_F \ell_F^2 \, \text{Tr}(\kappa_F)_i \right] + \alpha_C I_{\text{causal}}, \tag{7.2}$$

$$\text{Conscious: } \Phi_{\text{IFT}} > \Phi_c \wedge \tau \Xi > K. \tag{7.3}$$

Chapter 8

Final Declaration

IFT-Neuroscience transforms the theoretical framework of IFT-Consciousness into a falsifiable experimental program. Competitive experiments against IIT and GNWT, differentiable predictions for anesthesia and sleep, and real-time measurement protocols provide a clear roadmap for validating or refuting IFT in the laboratory.

$$\boxed{\text{IFT-Neuroscience} = \Phi_{\text{IFT}} \text{ measurable in real time}}.$$

$$\boxed{\text{Consciousness will be operational when } \Phi_c \text{ and } K \text{ are calibrated}}.$$

8.1 Affirmations

1. Φ_{IFT} is measurable with simultaneous EEG/fMRI.
2. IFT makes predictions differentiable from IIT and GNWT.
3. Anesthesia, sleep, and psychedelics affect distinct components of Φ_{IFT} and $\tau\Xi$.
4. Posterior cortex contributes more to Φ_{IFT} than frontal cortex for perceptual content.
5. The experimental program is falsifiable with preregistered criteria.